Credit Risk Portfolio Analytics — Solution Overview

Dashboard, view by view

- Portfolio Overview: exposure, accounts, overall default rate; default rate by any single dimension (loan type, employment, region, income band, age band); an employment-type x loan-type heatmap (cross-dimensional cut); 5 auto-generated insights with a "so what"; the data-quality decision log. An analyst uses this to spot which segments are driving losses and where to tighten policy.
- Account Explorer: pick any account_id (respecting the sidebar filters) to see its risk score, actual outcome, top SHAP drivers, repayment behaviour and bureau snapshot, plus global feature importance. An analyst uses this to explain a specific account's risk score to a credit committee.
- Ask the Portfolio (AI): free-text question box grounded in the currently filtered data.
- Anomaly Review (AI): ranked queue of flagged accounts with the reason for each flag and a Confirm/Dismiss control an analyst uses during periodic portfolio reviews.

Two AI features

- Ask the Portfolio: LLM (OpenAI, falls back to rule-based matching with no key) answers from a small pre-aggregated JSON context — never raw rows — and declines when the answer isn't in that context.
- Anomaly detection — trigger: Isolation Forest outlier (worst 5%) within an account's loan_type x employment_type peer segment on repayment/bureau behaviour features, OR a 30+ day DPD jump versus the start of the loan. False positive: a thin/new account with an atypical but benign history gets flagged. False negative: slow deterioration that stays near an already-risky segment's average is missed. Human-in-the-loop: analyst confirms (escalate) or dismisses each flag in the review queue; decisions are logged to artifacts/anomaly_decisions.csv.

Model results & approach

Target: target_default on the account grain (14,685 accounts, ~16.1% base rate). Features: customer demographics, loan terms, bureau rollups (utilization, DPD, write-offs) and repayment-history rollups (missed/late %, DPD trend, underpayment) aggregated per account. Baseline: Logistic Regression (class-balanced) — AUC 0.662, KS 0.244, Gini 0.323. Challenger: XGBoost — AUC 0.643, KS 0.207, Gini 0.286 (comparable discrimination, adds non-linear interactions and SHAP-based per-account explanations; LR is the more stable/interpretable production baseline here). Top drivers: loan_type=Mortgage, repay_avg_dpd, annual_income, repay_pct_late, employment_type=Unemployed.

Monitoring after deployment: track population stability index (PSI) on inputs and score distribution monthly, recalculate AUC/KS/Gini on each new vintage once defaults mature, alert on feature drift (e.g. bureau_utilization, repay_pct_missed), and retrain on a fixed cadence or when PSI exceeds a threshold.

Top 3 portfolio insights

- Unemployed borrowers default at ~35% vs ~13.6% for Business Owners — employment type should drive underwriting/pricing, not just rate.
- Mortgage is the riskiest product at ~24.7% default despite being secured, and carries the largest dollar exposure — pricing/collateral terms deserve review.
- Unemployed + Mortgage is a specific hotspot at ~54% default (n=70) — this combination needs its own rule rather than being averaged into broader policies.

Key assumptions & production needs

- Assumptions: synthetic data's missingness/labels mirror real messiness; a 2026-01-01 reference date for age-of-loan calcs; peer segments with <20 accounts are too small for stable anomaly scoring.
- From IT/data teams: a refreshed daily/weekly data feed (not static CSVs), a feature store or warehouse view for the bureau/repayment rollups, secrets management for the LLM key in production, and a model registry + approval workflow before any score is used for decisions.